

Dheeraj Reddy Mamidi

Applied AI / ML Engineer | LLM Applications | RAG | Predictive Modeling

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PROFESSIONAL SUMMARY

Applied AI/ML Engineer with 5+ years designing and shipping production LLM applications, RAG systems, and predictive models across education, consulting, and enterprise settings. Hands-on with Python, PyTorch, scikit-learn, FastAPI, Databricks, and Azure, delivering semantic search, agentic workflows, recommendation systems, and MLOps pipelines from concept to production. Known for translating ambiguous goals into responsible, measurable AI - lifting retrieval accuracy, cutting manual effort, and surfacing at-risk cases early.

TECHNICAL SKILLS

Generative AI & LLMs: LangChain, LangGraph, RAG, LlamaIndex, LLM Fine-tuning (LoRA, PEFT), Prompt Engineering, FAISS, Pinecone, Chroma, MCP, LangSmith

ML & Deep Learning: Scikit-learn, XGBoost, LightGBM, CatBoost, TensorFlow, PyTorch, LSTM, CNN, Transformers (BERT, GPT, LLaMA, Hugging Face), Isolation Forest

NLP & Explainability: spaCy, NLTK, Text Classification, Sentiment Analysis, NER, SHAP, LIME, RAGAS, Hybrid Retrieval (BM25 + vector)

Data Engineering: Apache Spark, PySpark, Kafka, Airflow, ETL/ELT, Databricks, AWS Glue, DVC, Feature Engineering, Schema Evolution

Cloud: AWS (S3, EC2, SageMaker, Lambda, Bedrock, Glue), Azure (ML, Databricks, Blob, Cognitive Services), GCP (Vertex AI)

MLOps & DevOps: MLflow, Kubeflow, Docker, Kubernetes (EKS/GKE), CI/CD, GitHub Actions, Jenkins, Terraform, Prometheus, Grafana, Evidently AI

Databases: PostgreSQL, MySQL, MongoDB, Neo4j, Snowflake, DynamoDB, Elasticsearch, Cosmos DB, SQL Server

Languages & Frameworks: Python (FastAPI, Flask, Django), SQL, R, Bash; REST APIs, Streamlit, Power BI, Tableau, Matplotlib, Seaborn

PROFESSIONAL EXPERIENCE

GenAI / ML Engineer

Jan 2025 - Present

Saint Louis University, St. Louis, MO

- Shipped an LLM-powered learning and research assistant adopted across multiple departments, cutting average time-to-answer for coursework and document queries by ~40% by combining RAG, semantic search, and role-based FastAPI services.
- Architected an enterprise RAG pipeline over research papers and institutional data using LangChain, LangGraph, and FAISS, raising answer relevance and grounding while cutting hallucinated responses.
- Lifted retrieval accuracy over keyword-only baselines with hybrid retrieval (BM25 + vector embeddings) across tens of thousands of educational documents.
- Orchestrated agentic multi-agent workflows in LangGraph to plan, retrieve, validate, and route tasks with human-in-the-loop review.
- Implemented responsible-AI guardrails (content filtering, prompt-injection checks, citation transparency, PII handling) and evaluation pipelines tracking retrieval accuracy, hallucination rate, and latency with LangSmith and RAGAS.
- Deployed containerized Docker services to production on Azure and Databricks with monitoring, structured logging, and reliable sub-second responses under load.

Data Scientist

Aug 2023 - Dec 2024

Boston Consulting Group (BCG), New York, NY

- Improved AUC-ROC ~14% over baseline and scored 500K+ daily records at sub-100ms latency by building classification and anomaly-detection models with XGBoost, LightGBM, and Isolation Forest for financial-services clients.

- Cut manual document review ~55% by delivering a BERT-based document-intelligence system that extracts and validates identity fields from scanned files.
- Satisfied regulatory review (EOCA/FCRA-aligned) by applying SHAP and LIME explainability and producing model cards and fairness analyses.
- Drove a ~14% lift in targeted product uptake via customer segmentation (K-Means, DBSCAN, hierarchical clustering) with silhouette validation.
- Managed the full model lifecycle with MLflow (tracking, registry, versioning) and automated drift detection; deployed models as REST services on AWS SageMaker.
- Built React dashboards backed by REST APIs for forecasts, anomaly alerts, and model metrics, and prototyped early LLM/RAG POCs for document search and summarization.

Data Engineer

Feb 2021 - Jun 2022

Bosch, Bangalore, India

- Centralized data from 6+ enterprise sources into a governed data lake by designing ETL/ELT pipelines in PySpark and Apache Airflow.
- Cut downstream pipeline errors ~45% and improved ML-input reliability with automated data-quality frameworks (schema checks, referential integrity, statistical outlier detection).
- Trimmed query time ~30% by modeling normalized PostgreSQL and MySQL schemas with tuned DDL, indexing, and stored procedures.
- Secured partner integrations with REST APIs using JWT, role-based access control, and CORS policies.
- Shortened release cycles with CI/CD and infrastructure-as-code; hardened releases with PyTest unit/integration tests and code-coverage gates.

PROJECTS

Personalized Learning-Pathway Recommender

Recommendation engine suggesting next-best learning modules from learner history and content embeddings, blending collaborative filtering with a RAG explanation layer; served via FastAPI on Databricks/Azure with feedback logging.

SignalForge AI - Personal Intelligence Capture Platform

Full-stack GenAI platform converting WhatsApp exports, URLs, files, screenshots, and audio/video into structured intelligence cards with classification, extracted questions, relevance scoring, and routing; reusable agent core, SSE streaming APIs, and a Next.js PWA.

CodeAtlas AI - Local RAG Code Intelligence Copilot

Local-first code understanding and review assistant that scans repos, redacts secrets, chunks code structurally, and retrieves via keyword/path/symbol/semantic search; grounded LLM reasoning, patch preview, a CLI, a local FastAPI daemon, and a VS Code extension scaffold.

RiskLens AI - Llama-Powered Exposure & Risk Analyzer

Domain-specific AI risk system using Llama to read risk/control/exposure data, classify findings, generate explainable risk summaries, and suggest remediation - applied GenAI for risk intelligence.

CERTIFICATIONS

- Databricks Certified Machine Learning Associate
- Microsoft Certified: Azure AI Engineer Associate (AI-102)
- AWS Certified Machine Learning - Specialty
- Databricks Certified Data Engineer Associate

EDUCATION

M.S., Information Systems - Southern Illinois University Edwardsville, IL

Aug 2022 - May 2023 | GPA 3.4 / 4.0

B.E., Computer Science - VJIT, Hyderabad, India

Jun 2017 - Apr 2021 | GPA 8.4 / 10